

Math 243: Homework 01

Due Wednesday, September 2

Problem 1 (Arc length). Let $f(x) = \frac{x^4}{8} + \frac{1}{4x^2}$. Compute the arc length of the curve from $x = 1$ to $x = 2$. Show your work.

Problem 2 (From practice midterm: Arc length). Compute the arc length of the parametrized curve:

$$\begin{cases} x(t) = \frac{4}{3}t^3 + t - 2 \\ y(t) = \frac{1}{27}(9t^2 + 2)^{\frac{3}{2}} \end{cases}$$

Problem 3. Find the length of the following curve in 3d space:

$$\mathbf{r}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} 1 \\ t^2 \\ t^3 \end{bmatrix}, \quad 0 \leq t \leq 9$$

Problem 4. Consider the following parametric equations

$$\begin{cases} x = 3 \cos(t) + 4 \\ y = \sin(t) + 2 \end{cases}$$

for all t in the interval $[0, 2\pi]$.

(a.) Fill in the following table values, and plot the points on a graph.

t	$x(t)$	$y(t)$
0		
$\pi/6$		
$\pi/4$		
$\pi/3$		
$\pi/2$		
$2\pi/3$		
$3\pi/4$		
$5\pi/6$		
π		

(b.) Convert the parametric equations to Cartesian form by eliminating the variable t . (*Hint: use the trig identity $\sin^2(t) + \cos^2(t) = 1$.*)

(c.) Sketch the graph of the curve as t ranges from 0 to 2π .